

LAB 6

CS23B096

Q1) switching from mtrap to user mode succesfully

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core 0: >>>> mtrap
core 0: 0x0000000080000050 (0x020043b7) lui      t2, 0x2004
(spike)
core 0: 0x0000000080000054 (0x0003ae03) lw      t3, 0(t2)
(spike)
core 0: 0x0000000080000058 (0x00006e89) c.lui    t4, 0x2
(spike)
core 0: 0x000000008000005a (0x710e8e9b) addiw    t4, t4, 1808
(spike)
core 0: 0x000000008000005e (0x00009e76) c.add    t3, t4
(spike)
core 0: 0x0000000080000060 (0x01c3a023) sw      t3, 0(t2)
(spike)
core 0: 0x0000000080000064 (0x30200073) mret
(spike)
core 0: >>>> user_mode
core 0: 0x0000000080000068 (0x00000013) nop
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Q2)

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core  0: >>>> context_switch
core  0: 0x0000000080000050 (0x00000f17) auipc    t5, 0x0
(spike)
core  0: 0x0000000080000054 (0x114f0f13) addi     t5, t5, 276
(spike)
core  0: 0x0000000080000058 (0x000f2f83) lw      t6, 0(t5)
(spike)
core  0: 0x000000008000005c (0x000f8463) beqz     t6, pc + 8
(spike)
core  0: >>>> save_context_A
core  0: 0x0000000080000064 (0x00000e97) auipc    t4, 0x0
(spike)
core  0: 0x0000000080000068 (0x0dce8e93) addi     t4, t4, 220
(spike)
core  0: 0x000000008000006c (0x005eb023) sd      t0, 0(t4)
(spike)
core  0: 0x0000000080000070 (0x34102f73) csrr     t5, mepc
(spike)
core  0: 0x0000000080000074 (0xffeebc23) sd      t5, -8(t4)
(spike)
core  0: 0x0000000080000078 (0x0000a835) c.j      pc + 60
(spike)
core  0: >>>> switch_to_B
core  0: 0x00000000800000b4 (0x00000397) auipc    t2, 0x0
(spike)
core  0: 0x00000000800000b8 (0x0ac38393) addi     t2, t2, 172
(spike)
core  0: 0x00000000800000bc (0x0003a383) lw      t2, 0(t2)
(spike)
core  0: 0x00000000800000c0 (0x02038463) beqz     t2, pc + 40
(spike)
core  0: >>>> initial_switch_to_B
core  0: 0x00000000800000e8 (0x00000e97) auipc    t4, 0x0
(spike)

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(spike)
core 0: >>>> initial_switch_to_B
core 0: 0x00000000800000e8 (0x00000e97) auipc    t4, 0x0
(spike)
core 0: 0x00000000800000ec (0x048e8e93) addi     t4, t4, 72
(spike)
core 0: 0x00000000800000f0 (0x341e9073) csrw     mepc, t4
(spike)
core 0: 0x00000000800000f4 (0x00004f05) c.li     t5, 1
(spike)
core 0: 0x00000000800000f6 (0x00000f97) auipc    t6, 0x0
(spike)
core 0: 0x00000000800000fa (0x06af8f93) addi     t6, t6, 106
(spike)
core 0: 0x00000000800000fe (0x01efa023) sw      t5, 0(t6)
(spike)
core 0: >>>> switch
core 0: 0x0000000080000102 (0x020043b7) lui     t2, 0x2004
(spike)
core 0: 0x0000000080000106 (0x2000ce37) lui     t3, 0x2000c
(spike)
core 0: 0x000000008000010a (0x00003e61) c.addiw  t3, -8
(spike)
core 0: 0x000000008000010c (0x000e2e83) lw      t4, 0(t3)
core 0: exception trap_load_access_fault, epc 0x000000008000010c
core 0:                tval 0x000000002000bfff8
(spike)
core 0: >>>> context_switch
core 0: 0x0000000080000050 (0x00000f17) auipc    t5, 0x0
(spike)
core 0: 0x0000000080000054 (0x114f0f13) addi     t5, t5, 276
(spike)
core 0: 0x0000000080000058 (0x000f2f83) lw      t6, 0(t5)
(spike)
core 0: 0x000000008000005c (0x000f8463) beqz     t6, pc + 8
(spike)
core 0: >>>> save_context_A
core 0: 0x0000000080000064 (0x00000e97) auipc    t4, 0x0
(spike) ☐

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(spike)
core  0: >>>> save_context_A
core  0: 0x0000000080000064 (0x00000e97) auipc    t4, 0x0
(spike)
core  0: 0x0000000080000068 (0x0dce8e93) addi     t4, t4, 220
(spike)
core  0: 0x000000008000006c (0x005eb023) sd      t0, 0(t4)
(spike)
core  0: 0x0000000080000070 (0x34102f73) csrr    t5, mepc
(spike)
core  0: 0x0000000080000074 (0xffeebc23) sd      t5, -8(t4)
(spike)
core  0: 0x0000000080000078 (0x0000a835) c.j      pc + 60
(spike)
core  0: >>>> switch_to_B
core  0: 0x00000000800000b4 (0x00000397) auipc    t2, 0x0
(spike)
core  0: 0x00000000800000b8 (0x0ac38393) addi     t2, t2, 172
(spike)
core  0: 0x00000000800000bc (0x0003a383) lw      t2, 0(t2)
(spike)
core  0: 0x00000000800000c0 (0x02038463) beqz    t2, pc + 40
(spike)
core  0: 0x00000000800000c4 (0x00000f17) auipc    t5, 0x0
(spike)
core  0: 0x00000000800000c8 (0x0a0f0f13) addi     t5, t5, 160
(spike)
core  0: 0x00000000800000cc (0x00004f85) c.li    t6, 1
(spike)
core  0: 0x00000000800000ce (0x01ff2023) sw      t6, 0(t5)
(spike)
core  0: 0x00000000800000d2 (0x00000e97) auipc    t4, 0x0
(spike)
core  0: 0x00000000800000d6 (0x07ee8e93) addi     t4, t4, 126
(spike)
core  0: 0x00000000800000da (0x000eb283) ld      t0, 0(t4)
(spike)
core  0: 0x00000000800000de (0xff8eb383) ld      t2, -8(t4)
(spike)

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